DocQA

Upload PDFs and ask questions about them. Answers cite the pages they come from. The LLM and embeddings run locally through Ollama.

```
docqa-backend/    FastAPI API and ARQ worker (Python 3.12), one shared image
docqa-frontend/   React + TypeScript + Vite
docs/             Architecture, backlog and story guides
docker-compose.yml
.env.example      Every environment variable, documented
```

Prerequisites

- Docker Desktop running (on Windows, with the WSL 2 backend)
- Git
- Ollama on the host machine. Nothing uses it yet; it is set up in S0.1.2.

You only need Python 3.12 and Node 22+ if you want to run the backend or frontend outside Docker.

Getting started

1. Create your local `.env` from the template:

   ```
   cp .env.example .env
   ```

   The defaults work for local development. `.env` is git-ignored, so never commit it.

2. Build and start the whole stack:

   ```
   docker compose up --build
   ```

   The first build takes a few minutes. Add `-d` to run it in the background.

3. Check that all five services are up, and that `postgres` and `redis` show as `healthy`:

   ```
   docker compose ps
   ```

Services

Service	URL / port	What it is
`frontend`	http://localhost:5173	Vite dev server with hot reload

Service	URL / port	What it is
`api`	http://localhost:8000/api/v1/health	FastAPI. It reloads automatically when you edit code.
	http://localhost:8000/docs	Interactive API docs
`worker`	(no port)	ARQ background worker. See its output with `docker compose logs -f worker`.
`postgres`	`localhost:5432`	PostgreSQL 16 with pgvector (user, password and database: `docqa`)
`redis`	`localhost:6379`	Redis 7, used as the job queue

The backend `app/` folder and the whole frontend folder are mounted into the containers, so code changes take effect without a rebuild. After you change `requirements.txt` or `package.json`, rebuild with `docker compose up --build`.

Stopping and resetting

```
docker compose down      # stop and remove containers; database data is kept
docker compose down -v   # also delete the volumes, which WIPES THE DATABASE
```

Uploaded files are stored in `./storage` on the host, and `down -v` does not touch them. Delete that folder by hand for a full reset.

Troubleshooting

Symptom	Fix
`port is already allocated`	Another process is using 5432, 6379, 8000 or 5173. Stop it, or change the left-hand port in `docker-compose.yml`.
Frontend doesn't hot-reload	`server.watch.usePolling` must stay `true` in `docqa-frontend/vite.config.ts`.
Frontend fails with missing packages after a dependency change	Run `docker compose up --build --renew-anon-volumes frontend`.
`api`/`worker` wait forever	Run `docker compose logs postgres redis` and check that the healthchecks pass.